

Selection of RF Test Instruments

Test instruments are usually very expensive. So proper care should be taken to select one that provides you the best value for your investment. To help you out, here we discuss the basic criteria for selecting the right RF measuring instrument for your needs



N432A power meter by Agilent

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Test instruments are used to make measurements. These are required in production units, R&D laboratories, calibration laboratories, maintenance and service centres, etc. For qualifying any product or subsystem of a product (such as amplifier, oscillator and transmitter) that comes out of a production unit, measurements are to be performed to find out whether the product meets the required specifications. After making measurements, one can declare that the units are working as per the given specifications and are ready for delivery. In R&D laboratories, instruments are needed to verify whether the developed module is behaving as per the design specifications.

You should judiciously select test instruments considering your present as well as future measurement requirements. As test instruments are usually very expensive, proper care should be taken to select one that provides you the best value for your investment.

To help you out, here we discuss the basic criteria for selecting the right RF measuring instrument for your needs.

General considerations

The most important and essential factor in selecting test instruments is their reliability. They should function within the given specifications at any time, in any place and over a long period. An unreliable instrument will give unreliable results, which is obviously not desired. Hence you should purchase these instruments from reputed vendors. The catalogue of the vendor should be studied for their selection of components, development and manufacturing stages. Vendors can be asked to supply the users' feedback on the performance of instruments.

The second important factor is the accuracy of the measurement claimed by the manufacturer. Accuracy is the closeness of the measured value to the true value. Since measuring instruments are mother tools, these should provide a high measurement accuracy. So ascertain the company's traceability

to national standards.

The third factor is after-sales and support facilities provided by the company. Even if the product is highly reliable, there is some component of unreliability associated with it. As per Murphy's Law, the instrument may fail at a crucial time. To reduce the downtime due to instrument failure, repair time is to be minimised. Prompt service by the company will naturally reduce the repair time. Hence the company should have a wide network of sales and service offices. The service office should be nearer to your location. For example, if the service centre is far away from your location, downtime will be more, affecting the production schedule. For the same reason, if the instrument is procured from abroad, the manufacturer should have service centres in India.

Technical considerations

Before purchasing an instrument, you should know what purpose the instrument is being purchased for and then generate specifications of the instrument. Specifications should take care of the current requirements as well as the possible future requirements to some extent. Over-specifications will unnecessarily increase the cost of the instrument.

For example, if an RF equipment manufacturer wants to purchase some test instrument, the instrument should cover all the frequency ranges in which the manufacturer is involved. The life of any test instrument is approximately 15 years. Beyond that period, the instrument will become obsolete and repair will not be economical. If you plan to increase the frequency range of your products in the next five years,